

Data Fusion Interview Challenge

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1 Statistical Setup

We consider two populations that both evolve over time $t = 1, 2, \dots$. The population over individuals in the general US population at time t is denoted by P_t , and the second is the the population over individuals in the US who respond to a survey at time t is denoted by S_t .

Let P_t be a joint distribution over an outcome $Y \in \mathcal{Y} \subset \mathbb{R}$ and a binary response indicator $R \in \{0, 1\}$ that corresponds to whether or not an individual responds to the survey. The outcome Y is a socioeconomic or health indicator, such as an individual's vaccination status, their employment status or their living standards. Let S_t be a distribution over outcomes conditional on respond to the survey $Y \mid R = 1$. This means that P_t is the joint distribution over (Y, R) , and S_t is the distribution over $Y \mid R = 1$. Note that the people who respond to the survey may not necessarily be representative of the US population, i.e. $S_t \neq P_t$.

We are interested in estimating the average outcome in the general population of the US at a current time step t . However, we only have access to timely but potentially non-representative data on the US population from surveys, i.e. we have access to S_t for only time steps $t = 1, 2, \dots M$, and representative but lagged data on the general US population, i.e. we have access to P_t for only for time steps $t = 1, 2, \dots m$ for $m < M$.

Our estimand of interest is

$$\mu(t) := \mathbb{E}_{P_t} [Y] \quad t = m + 1, \dots M.$$

Note that $\mu(t)$ for $t = m + 1, \dots M$ cannot be directly derived from our data because we do not have access to P_t when $t > m$.

Define the relative probability of sample selection between two individuals with outcome y and y' at time-step t as

$$w(y, y'; t) := \frac{\mathbb{P}_{P_t} [R = 1 \mid Y = y]}{\mathbb{P}_{P_t} [R = 1 \mid Y = y']}. \quad (1)$$

For example, suppose that Y corresponds to vaccination status, a binary variable. Then, we can omit the dependence of w on y, y' and define

$$w(t) := \frac{\mathbb{P}_{P_t} [R = 1 \mid Y = 1]}{\mathbb{P}_{P_t} [R = 1 \mid Y = 0]}.$$

When Y corresponds to vaccination status, the quantity $w(t)$ corresponds to how much more likely a vaccinated individual is to respond to survey than an unvaccinated individual.

We assume that that this relative probability is stable over time.

Assumption 1 (Stationary Selection). The function $w(y, y'; t)$ is constant in t .

In this note, we study to whether $\mu(t)$ can be identified, provided that Assumption 1 holds.

2 Algorithm

Let T be a random variable that denotes the time step and let Z be a pseudolabel that captures whether a sample is drawn from the survey or target population. Let $P_{t,Y}$ denote the marginal distribution over outcomes Y in P_t . The distribution F is a distribution over (T, Z, Y) , where

$$Z \sim \text{Bernoulli}(1/2), T \sim \text{Uniform}([m]), F_{Y|T=t, Z=1} = P_{t,Y}, F_{Y|T=t, Z=0} = S_t.$$

As a result, F can be viewed as a pooled distribution over the paired data from S_t, P_t for $t = 1, 2, \dots, m$.

Define a loss function

$$L(\theta, a, z) := (1 - z) \cdot \exp(\theta + a) - z \cdot (\theta + a). \quad (2)$$

We solve the augmented convex optimization problem

$$\{\tilde{\theta}(\cdot), \tilde{\alpha}(\cdot)\} \in \arg \min_{(\theta, \alpha) \in \Theta \times \mathcal{A}} \{\mathbb{E}_F [L(\theta(Y), \alpha(T), Z)]\}, \quad (3)$$

where Θ denotes a flexible function class (neural networks) and $\mathcal{A} \subset \mathbb{R}^m$ such that $\mathcal{A} = \{\alpha \in \mathbb{R}^m \mid \alpha(m) = 0\}$, where $\alpha(t)$ denotes the t -th index of vector α .

After solving the optimization problem in (3), we can obtain an estimator

$$\tilde{\mu}(t) := \mathbb{E}_{S_t} [\exp(\tilde{\theta}(Y)) \cdot Y] / \mathbb{E}_{S_t} [\exp(\tilde{\theta}(Y))].$$

3 Task

For $t = 1, \dots, m$, you observe independent samples from both the population outcome distribution P_t^Y and the survey distribution S_t . For $t = m + 1, \dots, M$, you observe samples only from S_t . Your goal is to use the historical paired data $t \leq m$ to learn a selection correction θ and use it to estimate $\tilde{\mu}$.

It may be helpful to skim [Sahoo et al. \[2022\]](#) Appendix Section B to see an example of a simple one-dimensional simulation.

Data-Generating Process

1. Let Y be a continuous-valued outcome. Specify distributions for S_t, P_t for $t = 1, 2, \dots, M$ that match the setting. Note that S_t, P_t can be defined arbitrarily but P_t should be a distribution over (Y, R) , S_t should be the $Y \mid R = 1$ -marginal of P_t , and P_t and S_t can change over time but Assumption 1 must hold. Ideally, $\mathbb{P}_{P_t} [R = 1]$ also changes over time. Create figures of S_t, P_t over time.
2. Write down a mathematical specification of the data generating process. How did you define P_t ? How did you define S_t ? How do you simulate drawing samples from S_t ? As an example, see Eq. 39 and Eq. 40 from [Sahoo et al. \[2022\]](#) Section B.
3. Draw n i.i.d. samples drawn from S_t for each $t = 1, 2, \dots, m$ and n i.i.d. samples drawn from P_t for each $t = 1, 2, \dots, m$. Create a dataset of i.i.d. samples from F (note that this dataset should have $2mn$ samples).

Optimization

1. Implement a Pytorch pipeline for learning $\tilde{\theta}, \tilde{\alpha}$ via (3) using your synthetic dataset from F . Hint: The set up of (θ, α) is closely related to the RU Regression parametrization from [Sahoo et al. \[2022\]](#).

2. Estimate $\tilde{\mu}(t)$ for $t = m + 1, \dots, M$ using your learned estimate of $\tilde{\theta}$ and the samples from S_t for $t > m$.

Conclusions

1. Compare $\mu(t)$ to $\tilde{\mu}(t)$ and to $\mathbb{E}_{S_t}[Y]$ for $t = m + 1 \dots M$ in a figure.

4 Deliverable

Please prepare concise three slides on your results and be prepared to talk through your implementation.

1. The first slide should capture the data-generating process in mathematical notation and in a figure.
2. The second slide should capture how you implemented solving the optimization problem.
3. The third slide should capture the comparison of $\mu(t), \mathbb{E}_{S_t}[Y], \tilde{\mu}(t)$ and any conclusions and learnings.

References

Roshni Sahoo, Lihua Lei, and Stefan Wager. Learning from a biased sample. *arXiv preprint arXiv:2209.01754*, 2022.